

COMPUTER NETWORKS

Part 3

Complete Study Notes

Unit 5 Transport Layer and Congestion Control

Unit 6 Application Layer and Wireless Networks

Plus Quick Revision & Formula Sheet

UNIT 5 — Transport Layer and Congestion Control

5.1 The Transport Layer — Role and Duties

The transport layer provides **process-to-process** (end-to-end) delivery. While the network layer delivers a packet to the correct *computer*, the transport layer delivers the message to the correct *process/application* running on that computer.

Responsibilities

1. **Process-to-process delivery** using **port numbers**
2. **Encapsulation and decapsulation** of messages into segments
3. **Segmentation and reassembly** with sequence numbers
4. **Connection control** — connectionless or connection-oriented
5. **Flow control** — end-to-end pacing
6. **Error control** — end-to-end reliability
7. **Multiplexing and demultiplexing** — many processes sharing one network connection
8. **Congestion control** — avoiding overloading the network

5.2 Addressing — Port Numbers

Port numbers are **16-bit integers (0 - 65,535)**.

Range	Name	Description
0 - 1023	Well-known ports	Assigned and controlled by IANA for standard services
1024 - 49151	Registered ports	Registered with IANA to prevent duplication; not controlled
49152 - 65535	Dynamic / Ephemeral ports	Temporary, assigned to client processes by the OS

Common Well-Known Ports

Port	Protocol	Service
20, 21	TCP	FTP (Data, Control)
22	TCP	SSH
23	TCP	Telnet
25	TCP	SMTP
53	TCP/UDP	DNS
67, 68	UDP	DHCP (Server, Client)
69	UDP	TFTP
80	TCP	HTTP
110	TCP	POP3
143	TCP	IMAP
161, 162	UDP	SNMP
443	TCP	HTTPS

Socket Address

A **socket address** is the combination of an **IP address + port number**, e.g. **192.168.1.5 : 8080**. A connection is uniquely identified by a **4-tuple**: (source IP, source port, destination IP, destination port).

5.3 UDP — User Datagram Protocol

UDP is a **connectionless, unreliable** transport protocol. It adds only port addressing and a checksum on top of IP — a "bare-bones" protocol.

UDP Header — 8 bytes, fixed

Field	Size	Purpose
Source Port	16 bits	Port of the sending process
Destination Port	16 bits	Port of the receiving process
Total Length	16 bits	Length of header + data (min 8, max 65,535)
Checksum	16 bits	Optional error detection over pseudoheader + header + data

Characteristics

- **No connection setup** — no handshake, so zero setup delay.
- **No acknowledgements, no retransmission, no sequencing.**
- **No flow control, no congestion control.**
- Preserves **message boundaries** (each write = one datagram).
- Supports **broadcast and multicast**.
- Very low overhead — only 8 bytes of header.

Where UDP is Used

- Simple request-response protocols: **DNS, DHCP, TFTP, SNMP, RIP, NTP**
- Real-time applications where late data is useless: **VoIP, video streaming, online gaming**
- Multicast/broadcast applications

UDP vs TCP

Feature	TCP	UDP
Connection	Connection-oriented	Connectionless
Reliability	Reliable (ACK + retransmission)	Unreliable, best-effort
Ordering	Guaranteed in-order delivery	No ordering guarantee
Header size	20–60 bytes	8 bytes
Flow control	Yes (sliding window)	No
Congestion control	Yes	No
Speed	Slower	Faster
Data boundary	Byte stream (no boundaries)	Message-oriented (preserves boundaries)
Broadcast/Multicast	No	Yes
Use cases	Web, email, file transfer	Streaming, gaming, DNS, VoIP

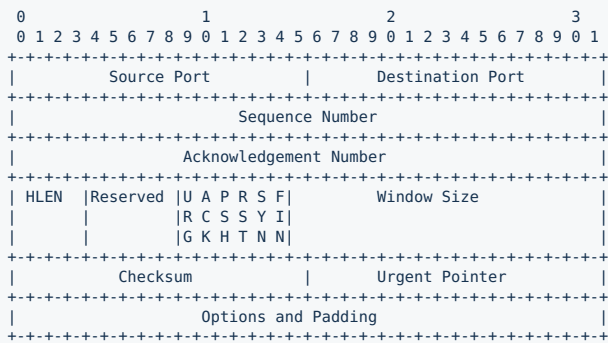
5.4 TCP — Transmission Control Protocol

TCP is a **connection-oriented, reliable, byte-stream** transport protocol.

Features

- **Stream delivery service** — treats data as a continuous stream of bytes, using sending and receiving **buffers**.
- **Full-duplex** communication.
- **Numbering system** — bytes are numbered, not segments:
- **Sequence Number** — number of the *first byte* of data carried in that segment.
- **Acknowledgement Number** — the number of the *next byte* the receiver expects (cumulative ACK).
- **Connection-oriented** — a virtual connection is established before data transfer.
- **Reliable** — uses ACKs, checksums, timers, and retransmission.
- **Flow control** and **congestion control**.

TCP Segment Header — 20 to 60 bytes



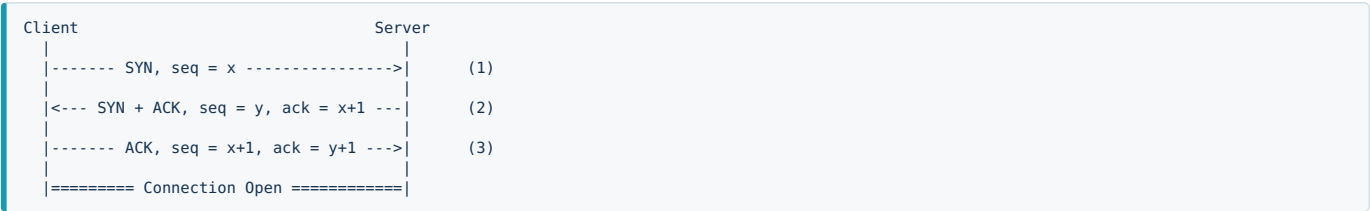
Field	Size	Purpose
Source Port	16 bits	Sending process port
Destination Port	16 bits	Receiving process port
Sequence Number	32 bits	Number of the first data byte in this segment
Acknowledgement Number	32 bits	Next byte expected from the other side
HLEN (Data Offset)	4 bits	Header length in 4-byte words (5 to 15)
Reserved	6 bits	For future use
Control Flags	6 bits	URG, ACK, PSH, RST, SYN, FIN
Window Size	16 bits	Receiver's advertised window (rwnd) in bytes — for flow control. Max 65,535
Checksum	16 bits	Mandatory error detection
Urgent Pointer	16 bits	Valid only when URG = 1; points to urgent data
Options	0–40 bytes	MSS, Window Scale, SACK Permitted, Timestamp

The Six Control Flags

Flag	Meaning
URG	The Urgent Pointer field is valid
ACK	The Acknowledgement Number field is valid
PSH	Push — deliver data to the application immediately, don't buffer
RST	Reset the connection (abort)
SYN	Synchronize sequence numbers during connection setup
FIN	Finish — sender has no more data (graceful close)

5.5 TCP Connection Management

Connection Establishment — Three-Way Handshake



- SYN** — the client sends a segment with SYN=1 and an initial sequence number (ISN) x. This segment carries no data but consumes one sequence number.
- SYN + ACK** — the server responds with SYN=1, ACK=1, its own ISN y, and ack = x+1. It also advertises its window size.
- ACK** — the client sends ACK=1 with ack = y+1. This segment may carry data.

SYN Flooding Attack: An attacker sends many SYN segments with spoofed source addresses. The server allocates resources (a half-open connection) for each and waits — eventually exhausting its connection table (a denial-of-service attack). **Defense:** SYN cookies, limiting half-open connections, filtering.

Data Transfer

After the handshake, both sides exchange data in full duplex, using sequence numbers, cumulative acknowledgements, and **piggybacking**.

Connection Termination — Four-Way Handshake

Client	Server	
----- FIN, seq = u ----->		(1)
<----- ACK, ack = u+1 -----		(2)
	(server sends remaining data)	
<----- FIN, seq = v -----		(3)
----- ACK, ack = v+1 ----->		(4)
	(TIME-WAIT: 2 × MSL)	

Half-Close: One side can stop sending while continuing to receive — useful when a client finishes sending a query but still awaits a long reply.

TCP State Transition (Key States)

CLOSED → LISTEN → SYN-SENT / SYN-RCVD → **ESTABLISHED** → FIN-WAIT-1 → FIN-WAIT-2 → TIME-WAIT → CLOSED (client side); CLOSE-WAIT → LAST-ACK → CLOSED (server side).

TIME-WAIT lasts **2 × MSL** (Maximum Segment Lifetime) to ensure the final ACK arrives and that old duplicate segments die out.

5.6 TCP Flow Control

Flow control prevents a **fast sender from overwhelming a slow receiver**.

TCP uses a **sliding window** protocol that is **byte-oriented** and **variable-size**.

- The receiver advertises **rwnd (receiver window)** — the amount of free buffer space — in the Window Size field of every segment.
- The sender must keep its unacknowledged data $\leq$ rwnd.

Sender window size = min(rwnd, cwnd) where **cwnd** is the congestion window (see below).

Silly Window Syndrome

Inefficiency where very small segments are exchanged, wasting bandwidth on headers.

- **Caused by the sender** (sending 1 byte at a time) → solved by **Nagle's Algorithm**: send the first byte immediately, then buffer subsequent bytes until either an ACK arrives or a full MSS accumulates.
- **Caused by the receiver** (advertising a tiny window) → solved by **Clark's Solution** (send an ACK with window = 0 until there is decent space) or **delayed acknowledgement**.

Zero-Window Probe

If the receiver advertises a window of 0, the sender periodically sends a 1-byte probe so it learns when the window reopens (preventing deadlock if the window-update ACK is lost).

5.7 TCP Error Control

TCP achieves reliability using three tools: **checksum, acknowledgement, and timeout**.

Acknowledgement Types

- **Cumulative ACK (ACK)** — acknowledges all bytes up to (but not including) the ack number.
- **Delayed ACK** — the receiver waits up to 500 ms (or until a second segment arrives) before acknowledging, to reduce traffic.
- **SACK (Selective Acknowledgement)** — an option letting the receiver report exactly which blocks it has received, so the sender retransmits only the missing pieces.

Retransmission

1. Retransmission After RTO (Retransmission Time-Out) Each sent segment starts a timer. If the timer expires before an ACK arrives, the segment is resent.

RTO calculation (Jacobson's algorithm):

RTT_smoothed = $(1 - \alpha) \times \text{RTT_smoothed} + \alpha \times \text{RTT_sample}$ ($\alpha \approx 0.125$) **RTT_deviation** = $(1 - \beta) \times \text{RTT_deviation} + \beta \times |\text{RTT_sample} - \text{RTT_smoothed}|$ ($\beta \approx 0.25$) **RTO** = **RTT_smoothed** + 4 × **RTT_deviation**

Karn's Algorithm: Do not use the RTT of a *retransmitted* segment to update the estimate (it is ambiguous which copy was acknowledged). Instead, apply **exponential back-off** — double the RTO on each retransmission.

2. Fast Retransmission (Three Duplicate ACKs) If the sender receives **three duplicate ACKs** for the same byte, it concludes the segment was lost and retransmits it **immediately**, without waiting for the timer.

5.8 Congestion and Congestion Control

What is Congestion?

Congestion occurs when the load on the network (the number of packets sent) exceeds the capacity of the network (the number it can handle). Routers' queues fill up, packets are dropped, delays skyrocket.

Effects of Congestion

- Increased **queuing delay**
- **Packet loss** due to buffer overflow
- Retransmissions, which add *more* load → a vicious cycle
- **Congestion collapse** — throughput drops sharply toward zero while the offered load keeps rising

Congestion Control vs Flow Control

Flow Control	Congestion Control
Protects the receiver from being overwhelmed	Protects the network from being overwhelmed
Between two endpoints	Involves all hosts and routers
Uses rwnd	Uses cwnd

Categories of Congestion Control

1. Open-Loop (Prevention) — policies applied *before* congestion happens. - Retransmission policy and timers - Window policy (Selective Repeat is better than Go-Back-N) - Acknowledgement policy - Discarding policy - Admission policy (in virtual-circuit networks)

2. Closed-Loop (Removal) — mechanisms that react *after* congestion is detected. - **Back pressure** — a congested node tells its upstream node to slow down; propagates backward (works only in virtual-circuit networks). - **Choke packet** — a router sends a packet directly to the source telling it to reduce rate (ICMP Source Quench). - **Implicit signaling** — the source *infers* congestion from delays/timeouts (this is what TCP does). - **Explicit signaling** — a router sets a bit in a packet to signal congestion, either **forward** (toward the destination, e.g. **ECN**) or **backward** (toward the source).

5.9 TCP Congestion Control Algorithms

TCP maintains a **congestion window (cwnd)** and a **slow start threshold (sssthresh)**. The actual send window = $\min(\text{rwnd}, \text{cwnd})$.

a) Slow Start (Exponential Increase)

- Start with **cwnd = 1 MSS**.
- For **every ACK received**, increase cwnd by **1 MSS**.
- Effectively, cwnd **doubles every RTT** → exponential growth: 1, 2, 4, 8, 16...
- Continues until **cwnd ≥ sssthresh** (then switch to Congestion Avoidance) or a loss is detected.

b) Congestion Avoidance (Additive Increase)

- Once cwnd reaches sssthresh, growth becomes **linear**.
- cwnd increases by **1 MSS per RTT** (i.e., by $\text{MSS} \times \text{MSS}/\text{cwnd}$ for each ACK).
- Growth: 16, 17, 18, 19...

c) Congestion Detection (Multiplicative Decrease)

Case 1 — Timeout (severe congestion assumed): - $\text{sssthresh} = \text{cwnd} / 2$ - **cwnd = 1 MSS** - Restart **Slow Start**

Case 2 — Three duplicate ACKs (mild congestion, network still delivering): - $\text{sssthresh} = \text{cwnd} / 2$ - **cwnd = sssthresh** (in TCP Reno: $\text{cwnd} = \text{sssthresh} + 3 \text{ MSS}$) - Enter **Fast Recovery**, then continue in Congestion Avoidance

This overall behavior is called **AIMD — Additive Increase, Multiplicative Decrease** and produces TCP's characteristic "sawtooth" throughput graph.

TCP Variants

Variant	Key characteristic
TCP Tahoe	Slow Start + Congestion Avoidance + Fast Retransmit. Any loss → cwnd = 1
TCP Reno	Adds Fast Recovery — on 3 dup ACKs, halve cwnd instead of dropping to 1
TCP New Reno	Improves Reno's handling of multiple losses in one window
TCP SACK	Uses selective acknowledgements to retransmit only missing blocks
TCP Vegas	Detects congestion from <i>increasing RTT</i> before loss occurs (delay-based)
TCP CUBIC	Window growth is a cubic function of time since last loss; default in Linux, good for high-bandwidth long-delay links
BBR	Models bottleneck bandwidth and RTT rather than reacting to loss (Google)

5.10 Quality of Service (QoS)

QoS is a set of techniques to give different traffic flows the treatment they need.

Flow Characteristics

Characteristic	Meaning
Reliability	Tolerance to packet loss (email needs high; VoIP tolerates some)
Delay	Latency tolerance (video conferencing needs low)
Jitter	Variation in delay (critical for real-time audio/video)
Bandwidth	Data rate required (video needs the most)

Techniques to Improve QoS

1. Scheduling - FIFO Queuing — packets served in arrival order; simple but no differentiation. - **Priority Queuing** — multiple queues by priority; higher-priority queues served first. Risk: **starvation** of lower priorities. - **Weighted Fair Queuing (WFQ)** — each queue gets a weight; the scheduler serves queues in proportion to their weights. Avoids starvation.

2. Traffic Shaping - Leaky Bucket — packets are queued and output at a **constant rate**, smoothing bursts. Excess packets overflow and are discarded. Very rigid. - **Token Bucket** — tokens accumulate in a bucket at a fixed rate; a packet may be sent only if a token is available. **Allows bursts** up to the bucket size — more flexible than leaky bucket and idle hosts get credit. - Often combined: token bucket for burst allowance, leaky bucket for rate limiting.

3. Resource Reservation — buffers, bandwidth, and CPU time reserved in advance (RSVP).

4. Admission Control — a router accepts a new flow only if it has the resources.

QoS Models

a) Integrated Services (IntServ) — flow-based; uses **RSVP (Resource Reservation Protocol)** to reserve resources per flow. Offers Guaranteed Service and Controlled-Load Service. *Problem:* Poor scalability (per-flow state in every router).

b) Differentiated Services (DiffServ) — class-based. Packets are marked with a **DSCP** value in the IP header; routers apply a **Per-Hop Behavior (PHB)** per class. Much more scalable — no per-flow state. PHBs: **EF (Expedited Forwarding)**, **AF (Assured Forwarding)**, **Default (best effort)**.

5.11 SCTP — Stream Control Transmission Protocol (Brief)

A newer reliable, message-oriented transport protocol that combines the best of TCP and UDP.

Key features: - **Multistreaming** — multiple independent streams within one association, so a loss in one stream doesn't block others (avoids TCP's **head-of-line blocking**). - **Multihoming** — an association can use multiple IP addresses per endpoint for fault tolerance. - **Message-oriented** (preserves boundaries like UDP) but **reliable and ordered** (like TCP). - **Four-way handshake with cookies** — immune to SYN flooding attacks. - Used in telephony signaling (SIGTRAN), and as the transport for WebRTC data channels.

UNIT 6 — Application Layer and Wireless Networks

PART A — The Application Layer

6.1 Application Layer Paradigms

a) Client-Server Paradigm

- The **server** runs continuously, waiting for requests; the **client** initiates requests.
- Server has a fixed, well-known address.
- Examples: HTTP/Web, FTP, email, DNS.
- *Limitation*: server can become a bottleneck and a single point of failure.

b) Peer-to-Peer (P2P) Paradigm

- No always-on server; peers communicate directly and act as both client and server.
- **Self-scaling** — each new peer adds capacity as well as demand.
- Types: **Centralized** (a directory server tracks peers, e.g. original Napster), **Decentralized/unstructured** (flooding queries, e.g. Gnutella), **Structured with DHT (Distributed Hash Table)** (e.g. Chord, Pastry, Kademlia).
- Example: **BitTorrent** — files split into chunks, "tit-for-tat" incentives, tracker coordinates a swarm.

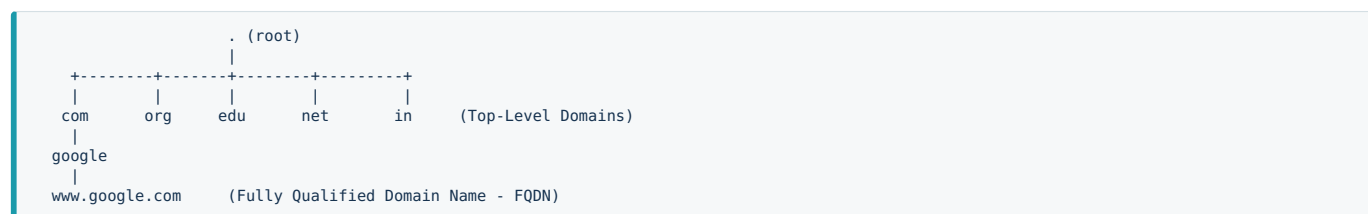
c) Hybrid Paradigm

Combines both — e.g., Skype (originally) used central servers for login/directory, and P2P for the actual media.

6.2 DNS — Domain Name System

DNS maps human-readable **domain names** to **IP addresses**. It is a **distributed, hierarchical database**.

Name Space Hierarchy



- **Root domain** — represented by a dot.
- **TLDs (Top-Level Domains)**:
- **gTLD (generic)** — .com, .org, .net, .edu, .gov, .mil, .int
- **ccTLD (country code)** — .in, .uk, .jp, .us
- **Second-level domains** — registered by organizations.
- **Subdomains** — e.g. `mail.google.com`.

DNS Server Types

Type	Role
Root server	Knows the authoritative servers for each TLD (13 logical root server clusters worldwide)
TLD server	Knows authoritative servers for domains under its TLD
Authoritative server	Holds the actual records for a specific zone/domain
Local / Recursive resolver	The server your ISP/network provides; performs lookups on your behalf and caches results

Resolution Types

- **Recursive resolution** — the client asks its local resolver for the final answer; the resolver does all the work and returns the complete answer.
- **Iterative resolution** — each server returns a referral ("ask this server instead"), and the resolver follows the chain itself.

Common DNS Record Types

Record	Purpose
A	Maps a hostname to an IPv4 address
AAAA	Maps a hostname to an IPv6 address
CNAME	Canonical name — an alias pointing to another name
MX	Mail exchange — mail server for the domain (with priority)
NS	Name server for the zone
PTR	Pointer — reverse lookup (IP → name)
SOA	Start of Authority — zone administrative info
TXT	Arbitrary text (SPF, DKIM, domain verification)
SRV	Service location record

Other Points

- DNS uses **UDP port 53** for queries (fast, low overhead) and **TCP port 53** for zone transfers and responses larger than 512 bytes.
- Caching** with a **TTL** reduces load and latency.
- DNSSEC** adds cryptographic signatures to protect against cache poisoning/spoofing.

6.3 HTTP — HyperText Transfer Protocol

The protocol of the **World Wide Web**. It is a **stateless**, request-response protocol running over **TCP port 80** (HTTPS uses TLS over port 443).

HTTP Message Format

Request message:

```
GET /index.html HTTP/1.1      ← Request line (Method, URL, Version)
Host: www.example.com        ← Headers
User-Agent: Mozilla/5.0
Accept: text/html
Connection: keep-alive

[optional body]              ← Blank line
```

Response message:

```
HTTP/1.1 200 OK               ← Status line (Version, Status code, Phrase)
Date: Wed, 16 Sep 2026 10:00:00 GMT
Server: nginx
Content-Type: text/html
Content-Length: 1234

<html>...</html>              ← Blank line
                                ← Body
```

HTTP Methods

Method	Purpose
GET	Retrieve a resource
POST	Submit data to be processed; creates/updates
PUT	Store/replace a resource at the given URL
DELETE	Remove a resource
HEAD	Like GET but returns headers only
OPTIONS	Ask what methods the server supports
TRACE	Echo the request for diagnostics
CONNECT	Establish a tunnel (used for HTTPS through proxies)

Status Code Classes

Range	Class	Examples
1xx	Informational	100 Continue, 101 Switching Protocols
2xx	Success	200 OK, 201 Created, 204 No Content
3xx	Redirection	301 Moved Permanently, 302 Found, 304 Not Modified
4xx	Client error	400 Bad Request, 401 Unauthorized, 403 Forbidden, 404 Not Found
5xx	Server error	500 Internal Server Error, 502 Bad Gateway, 503 Service Unavailable

Connection Types

- **Non-persistent (HTTP/1.0)** — a new TCP connection for every object. Requires 2 RTTs per object plus slow-start penalty.
- **Persistent (HTTP/1.1, default)** — one TCP connection carries multiple requests/responses. With **pipelining**, requests can be sent back to back.

Statelessness and Cookies

HTTP itself keeps no state between requests. **Cookies** (Set-Cookie header from the server, Cookie header from the client) allow sessions, authentication, personalization, and tracking.

Proxy Servers and Caching

A **proxy server** stores copies of recent responses. Benefits: reduced latency, reduced external traffic, and access control.

HTTP Versions

- **HTTP/1.0** — non-persistent, one request per connection.
- **HTTP/1.1** — persistent connections, pipelining, host header, chunked transfer.
- **HTTP/2** — binary framing, true multiplexing over one connection, header compression (HPACK), server push.
- **HTTP/3** — runs over **QUIC (UDP-based)**, eliminating TCP head-of-line blocking; faster connection setup.

HTTPS

HTTP over **TLS/SSL**. Provides confidentiality (encryption), integrity, and authentication (via certificates from a Certificate Authority).

6.4 FTP — File Transfer Protocol

Transfers files between hosts. Notable for using **two separate TCP connections**:

- **Control connection — port 21** — stays open for the whole session; carries commands and replies (using NVT ASCII).
- **Data connection — port 20** — opened and closed for each file transfer.

Modes

- **Active mode** — the server initiates the data connection back to the client (problematic with firewalls/NAT).
- **Passive mode** — the client initiates both connections; the server just tells the client which port to connect to. This is the common mode today.

File transfer attributes

- **File type**: ASCII, EBCDIC, Image (binary)
- **Data structure**: file, record, page
- **Transmission mode**: stream, block, compressed

Anonymous FTP allows public access with the username `anonymous`.

Security note: FTP sends credentials in **plaintext**. Secure alternatives: **SFTP** (over SSH, port 22), **FTPS** (FTP over TLS).

TFTP (Trivial FTP) — a stripped-down version over **UDP port 69**, no authentication; used for booting diskless devices and loading router firmware.

6.5 Electronic Mail

Architecture Components

- **UA (User Agent)** — the mail client (Outlook, Thunderbird, Gmail interface). Composes, reads, replies, forwards.
- **MTA (Message Transfer Agent)** — the mail server that relays messages (e.g. Sendmail, Postfix).
- **MAA (Message Access Agent)** — lets the recipient retrieve mail from the mailbox.

Flow

Sender UA → (SMTP) → Sender's Mail Server → (SMTP) → Receiver's Mail Server → (POP3/IMAP) → Receiver UA

SMTP — Simple Mail Transfer Protocol

- **TCP port 25** (587 for submission with authentication, 465 for SMTPS).
- A **push** protocol — used to *send* mail.
- Commands: HELO/EHLO, MAIL FROM, RCPT TO, DATA, QUIT, RSET, VRFY.
- Responses are 3-digit codes (e.g., 220 Service ready, 250 OK, 354 Start mail input, 550 Mailbox unavailable).
- Limitation: originally handled only 7-bit NVT ASCII.

MIME — Multipurpose Internet Mail Extensions

An extension (not a separate protocol) that lets non-ASCII data be sent via SMTP by transforming it to ASCII at the sender and

back at the receiver. - **MIME headers:** MIME-Version, Content-Type, Content-Transfer-Encoding, Content-ID, Content-Description. - **Content types:** text/plain, text/html, image/jpeg, audio/basic, video/mpeg, application/pdf, multipart/mixed. - **Encodings:** 7bit, 8bit, binary, **Base64**, **Quoted-printable**.

POP3 vs IMAP

Feature	POP3 (port 110)	IMAP4 (port 143)
Model	Download and (usually) delete	Mail stays on the server
Multiple devices	Poor support	Excellent — synchronized across devices
Folders on server	No	Yes
Partial download	No — full message only	Yes — can fetch headers first
Search on server	No	Yes
Offline access	Good (fully downloaded)	Requires syncing
Complexity / storage	Simple, less server storage	Complex, more server storage
Secure ports	995 (POP3S)	993 (IMAPS)

Web-Based Mail

The transfer between the user's browser and the mail server uses **HTTP**; SMTP is still used between mail servers.

6.6 DHCP — Dynamic Host Configuration Protocol

DHCP automatically assigns network configuration to hosts: **IP address, subnet mask, default gateway, and DNS server addresses**.

- Runs over **UDP** — server port **67**, client port **68**.
- Successor to BOOTP; backward compatible.

The DORA Process

Step	Message	Description
D	DHCP Discover	Client broadcasts to find any DHCP server (src 0.0.0.0, dst 255.255.255.255)
O	DHCP Offer	Server(s) offer an IP address and configuration
R	DHCP Request	Client broadcasts acceptance of one offer
A	DHCP Acknowledgement	Server confirms and commits the lease

Lease time: addresses are leased temporarily. The client attempts renewal at **T1 = 50%** of the lease and rebinding at **T2 = 87.5%**.

Other messages: DHCP Release, DHCP Decline, DHCP NAK, DHCP Inform.

DHCP Relay Agent: Because DHCP uses broadcasts (which routers block), a relay agent forwards DHCP messages to a server on another subnet.

6.7 Other Application Layer Protocols

TELNET

Remote terminal login (**TCP port 23**) using the **NVT (Network Virtual Terminal)** character set. **Insecure** — transmits everything, including passwords, in plaintext.

SSH — Secure Shell

The secure replacement for Telnet, rlogin, and FTP (**TCP port 22**). Provides encryption, server authentication, and integrity. Components: SSH-TRANS (transport, secure channel), SSH-AUTH (authentication), SSH-CONN (connection multiplexing). Also supports **port forwarding / tunneling**.

SNMP — Simple Network Management Protocol

Used to monitor and manage network devices (**UDP ports 161 for agents, 162 for traps**). - **Manager** — the monitoring station. - **Agent** — software on the managed device (router, switch, server). - **MIB (Management Information Base)** — the database of objects that can be managed. - **SMI (Structure of Management Information)** — rules for naming and defining object types. - **Messages:** GetRequest, GetNextRequest, GetBulkRequest, SetRequest, Response, **Trap** (agent-initiated alert), InformRequest.

NTP — Network Time Protocol

Synchronizes clocks across devices (**UDP port 123**), using a hierarchy of **stratum** levels (stratum 0 = atomic clocks/GPS).

WWW — World Wide Web

A distributed client-server hypermedia service. - **URL (Uniform Resource Locator)** structure: `protocol://host:port/path?query#fragment` - **Web documents:** Static (HTML, fixed content), Dynamic (generated on the server per request — PHP, JSP, CGI), Active (code runs in the client's browser — JavaScript, Java applets historically).

PART B — Wireless Networks

6.8 Wireless vs Wired — Key Differences

Aspect	Wired	Wireless
Medium	Cable (guided)	Air / free space (unguided)
Mobility	None	High
Bandwidth	High and stable	Lower, variable
Error rate	Very low	High (interference, fading, multipath)
Security	Physically contained	Broadcast — anyone in range can listen
Collision detection	Possible (CSMA/CD)	Not possible → CSMA/CA used
Installation cost	High (cabling)	Low

Wireless Transmission Problems

- **Attenuation** — signal weakens rapidly with distance (inverse square law).
- **Interference** — from other devices in the same band (microwaves, Bluetooth).
- **Multipath propagation** — signal reaches the receiver by multiple reflected paths, causing fading and distortion.
- **Hidden terminal problem** — A and C cannot hear each other but both can reach B; they collide at B. → Solved by RTS/CTS.
- **Exposed terminal problem** — A station refrains from transmitting because it senses a transmission that would not actually interfere with its intended receiver, wasting capacity.

6.9 IEEE 802.11 — Wireless LAN (Wi-Fi)

Architecture

BSS (Basic Service Set) — the building block. - **Infrastructure BSS** — stations connect through an **AP (Access Point)**. - **IBSS (Independent BSS) / Ad hoc network** — stations communicate directly without an AP.

ESS (Extended Service Set) — two or more BSSs joined by a **distribution system** (usually wired Ethernet). Allows roaming.

Station types: No-transition (stationary or moving within a BSS), BSS-transition (moving between BSSs in one ESS), ESS-transition (moving between ESSs).

MAC Sublayer

- **DCF (Distributed Coordination Function)** — the standard contention-based access using **CSMA/CA** (see Unit 3). Mandatory.
- **PCF (Point Coordination Function)** — optional, centralized, contention-free polling by the AP. Rarely implemented.

Interframe Spaces: SIFS (shortest, for ACK/CTS) < PIFS (PCF) < DIFS (DCF) — the shorter the IFS, the higher the priority.

Frame Format (simplified)

Frame Control | Duration/ID | Address 1 | Address 2 | Address 3 | Sequence Control | Address 4 | Frame Body | FCS

Three (sometimes four) address fields are needed because a frame may travel via an AP and a distribution system. The **To DS** / **From DS** bits in the Frame Control field determine how the addresses are interpreted.

Frame types: Management (association, authentication, beacon, probe), Control (RTS, CTS, ACK), Data.

802.11 Standards

Standard	Marketing name	Band	Max data rate	Notes
802.11 (legacy)	—	2.4 GHz	2 Mbps	Original, 1997
802.11a	—	5 GHz	54 Mbps	OFDM; shorter range
802.11b	—	2.4 GHz	11 Mbps	DSSS; long range, crowded band
802.11g	—	2.4 GHz	54 Mbps	Backward compatible with b
802.11n	Wi-Fi 4	2.4 / 5 GHz	600 Mbps	Introduced MIMO , channel bonding
802.11ac	Wi-Fi 5	5 GHz	~3.5 Gbps	MU-MIMO, 80/160 MHz channels
802.11ax	Wi-Fi 6 / 6E	2.4 / 5 / 6 GHz	~9.6 Gbps	OFDMA , better efficiency in dense areas, TWT
802.11be	Wi-Fi 7	2.4 / 5 / 6 GHz	~46 Gbps	320 MHz channels, Multi-Link Operation, 4K-QAM

Wireless Security

Protocol	Year	Encryption	Status
WEP	1999	RC4, 40/104-bit key	Broken — trivially cracked
WPA	2003	TKIP + RC4	Interim fix; now deprecated
WPA2	2004	AES-CCMP	Long-time standard; vulnerable to KRACK
WPA3	2018	AES-GCMP, SAE handshake	Current; forward secrecy, protects weak passwords

Modes: Personal (PSK — pre-shared key) and Enterprise (802.1X/RADIUS authentication).

6.10 Bluetooth — IEEE 802.15

A short-range wireless technology for **WPANs**, designed to replace cables.

Architecture

- **Piconet** — one **master** and up to **7 active slaves** (plus up to 255 parked slaves). All devices are synchronized to the master's clock and hopping sequence.
- **Scatternet** — multiple piconets connected; a slave in one piconet can be the master in another.

Technical Details

- Operates in the **2.4 GHz ISM band**.
- Uses **FHSS (Frequency Hopping Spread Spectrum)** — 79 channels, **1600 hops per second**.
- **TDMA-based** time slots of 625 μ s; the master controls all communication.
- **Classes:** Class 1 (100 m), Class 2 (10 m — most common), Class 3 (1 m).

Protocol Stack Layers

Radio → Baseband → **L2CAP (Logical Link Control and Adaptation Protocol)** → Upper layers/profiles.

Link types: **SCO (Synchronous Connection-Oriented)** for real-time voice (no retransmission); **ACL (Asynchronous Connectionless Link)** for data (retransmission supported).

BLE — Bluetooth Low Energy

Introduced in Bluetooth 4.0 for IoT: very low power, 40 channels, ideal for wearables, beacons, and sensors that run for months on a coin cell.

6.11 Cellular Networks

Basic Concepts

- The service area is divided into **cells**, each with a **Base Station (BS)**.
- **Frequency reuse** — the same frequencies are reused in non-adjacent cells, separated enough to avoid interference. This is what gives cellular systems their capacity.
- **MSC (Mobile Switching Center)** connects cells to each other and to the PSTN, and tracks subscriber location.
- **Handoff / Handover** — transferring an ongoing call from one cell to another as the user moves.
- **Hard handoff** — break the old connection before making the new one ("break before make"). Used in GSM.
- **Soft handoff** — connect to the new cell before releasing the old one ("make before break"). Used in CDMA.
- **Roaming** — using the network of a different operator outside the home area.

Generations

Gen	Era	Technology	Key capability
1G	1980s	AMPS — Analog, FDMA	Voice only, poor quality and security
2G	1990s	GSM (TDMA/FDMA), IS-95 (CDMA)	Digital voice, SMS ; 2.5G added GPRS , 2.75G added EDGE
3G	2000s	UMTS/WCDMA, CDMA2000, HSPA	Mobile Internet, video calling; up to a few Mbps
4G	2010s	LTE / LTE-Advanced , OFDMA, MIMO	All-IP network, up to 1 Gbps, HD streaming, VoLTE
5G	2020s	NR (New Radio), mmWave, massive MIMO, beamforming	Up to 10–20 Gbps, ultra-low latency (~1 ms) , massive IoT density, network slicing

5G service categories: **eMBB** (enhanced Mobile Broadband), **URLLC** (Ultra-Reliable Low-Latency Communication), **mmTC** (massive Machine-Type Communication).

6.12 WiMAX — IEEE 802.16

Worldwide Interoperability for Microwave Access — a standard for **Wireless MAN** / broadband wireless access, sometimes called "last-mile wireless broadband".

- Range of up to **50 km** (fixed) and data rates up to ~70 Mbps.
- Supports both **line-of-sight** (higher frequency, longer range) and **non-line-of-sight** operation.
- Connection-oriented MAC with built-in QoS classes.
- Largely superseded by LTE in practice.

6.13 Satellite Networks

Orbit	Altitude	Propagation delay	Satellites needed for global coverage	Examples
GEO (Geostationary)	35,786 km	~250 ms one way	3	TV broadcast, weather
MEO (Medium Earth Orbit)	5,000 – 15,000 km	~50–150 ms	~10–20	GPS , Galileo
LEO (Low Earth Orbit)	500 – 2,000 km	~5–10 ms	40–80+ per constellation	Iridium, Starlink

GEO advantage: appears stationary from earth, so antennas need not track it. **LEO advantage:** low latency and low power, but requires large constellations and handoff between satellites.

6.14 Mobile IP

Allows a mobile host to keep its **permanent IP address** while moving between networks, so existing connections survive.

Key Entities and Addresses

- **Home Address** — the mobile node's permanent IP address.
- **Home Agent (HA)** — a router on the home network that maintains the binding and intercepts packets.
- **Foreign Agent (FA)** — a router on the visited network.
- **Care-of Address (COA)** — the temporary address on the foreign network.

Three Phases

1. **Agent Discovery** — the mobile node learns whether it is on its home or a foreign network (via agent advertisement/solicitation).
2. **Registration** — the mobile node registers its care-of address with its home agent.
3. **Data Transfer** — packets sent to the home address are intercepted by the HA, **tunneled** (encapsulated) to the COA, and delivered by the FA.

Inefficiencies

- **Triangle routing (Double crossing)** — packets travel a long way from sender → home network → foreign network even when the sender is close by. Mitigated by **route optimization** (informing the correspondent node of the COA directly).

Quick Revision — Master Summary

Layer Cheat Sheet

Layer	Unit of Data	Key Functions	Devices	Key Protocols
Application	Message	User services	—	HTTP, FTP, SMTP, DNS, DHCP, SNMP
Presentation	—	Translation, encryption, compression	—	SSL/TLS, JPEG, MPEG
Session	—	Dialog control, synchronization	—	NetBIOS, RPC
Transport	Segment / Datagram	Process delivery, reliability, flow & congestion control	Gateway	TCP, UDP, SCTP
Network	Packet	Logical addressing, routing	Router	IP, ICMP, IGMP, ARP, OSPF, BGP
Data Link	Frame	Framing, MAC addressing, error control	Switch, Bridge, NIC	Ethernet, PPP, HDLC, 802.11
Physical	Bit	Signals, media, encoding	Hub, Repeater, Cable	10Base-T, DSL, RS-232

Formulas to Remember

Concept	Formula
Nyquist bit rate	$\text{BitRate} = 2 \times B \times \log_2(L)$
Shannon capacity	$C = B \times \log_2(1 + \text{SNR})$
dB	$\text{dB} = 10 \log_{10} (P_2/P_1)$
Wavelength	$\lambda = c / f$
Period ↔ Frequency	$T = 1/f$
Sampling rate	$f_s \geq 2 \times f_{\text{max}}$
Mesh links	$n(n-1)/2$
Hamming redundancy	$2^r \geq m + r + 1$
Addresses in a /n block	$2^{(32-n)}$
Usable hosts	$2^{(32-n)} - 2$
Go-Back-N window	$\leq 2^m - 1$
Selective Repeat window	$2^{(m-1)}$
Pure ALOHA throughput	$S = G \cdot e^{(-2G)}$, max 18.4%
Slotted ALOHA throughput	$S = G \cdot e^{(-G)}$, max 36.8%
TCP send window	$\min(\text{rwnd}, \text{cwnd})$
RTO	$\text{RTT}_s + 4 \times \text{RTT}_d$
OSPF cost	$10^8 / \text{bandwidth}(\text{bps})$

Port Numbers to Memorize

20/21 FTP · 22 SSH · 23 Telnet · 25 SMTP · 53 DNS · 67/68 DHCP · 69 TFTP · 80 HTTP · 110 POP3 · 143 IMAP · 161/162 SNMP · 179 BGP · 443 HTTPS · 520 RIP

Protocol Numbers (in IP header)

1 ICMP · 2 IGMP · 6 TCP · 17 UDP · 89 OSPF

End of the Computer Networks study book — all six units covered.